



**KNOX GRAMMAR SCHOOL**

**Name:** \_\_\_\_\_

**Teacher:** \_\_\_\_\_

**2020**

**Year 11 Yearly Examination**

# **Mathematics Extension 1**

## **General Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

## **Teachers:**

**Mr Cheah**

**Mr Lemaire**

**Mr Meli**

**Mr Willcocks**

**Ms Yun**

## **Total marks: 70**

**Section I: 10 marks** (pages 3 – 6)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

**Section II: 60 marks** (pages 7 – 12)

- Attempt Questions 11 – 14
- Allow about 1 hour 45 minutes for this section

**Setter: YUS**

**This paper MUST NOT be removed from the examination room**

**Write your full name and your teacher's name on the front cover of each writing booklet**

**Number of Students in Course: 107**

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## Section I

10 marks

Attempt questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

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- 1 It is given that  $\log_a 8 = 1.893$ , correct to 3 decimal places.

What is the value of  $\log_a 4$ , correct to 2 decimal places?

- A. 0.95
- B. 1.26
- C. 1.53
- D. 2.84

- 2 What is the domain of the function  $f(x) = \sqrt{x^2 - 3x}$ ?

- A.  $x \geq 3$
- B.  $x > 3$
- C.  $x \leq 0$  or  $x \geq 3$
- D.  $x < 0$  or  $x > 3$

- 3 A polynomial equation has roots  $\alpha$ ,  $\beta$  and  $\gamma$  where

$$\alpha + \beta + \gamma = -2, \alpha\beta + \alpha\gamma + \beta\gamma = 3 \text{ and } \alpha\beta\gamma = 1.$$

Which polynomial equation has the roots  $\alpha$ ,  $\beta$  and  $\gamma$ ?

- A.  $x^3 + 2x^2 + 3x + 1 = 0$
- B.  $x^3 + 2x^2 + 3x - 1 = 0$
- C.  $x^3 - 2x^2 + 3x + 1 = 0$
- D.  $x^3 - 2x^2 + 3x - 1 = 0$

4 Which expression is the equivalent to  $\sin 2x \cos 2x$ ?

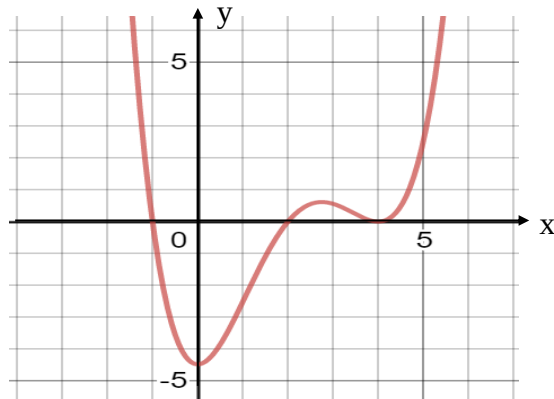
A.  $\frac{1}{2} \sin 4x$

B.  $2 \sin 4x$

C.  $\sin 4x$

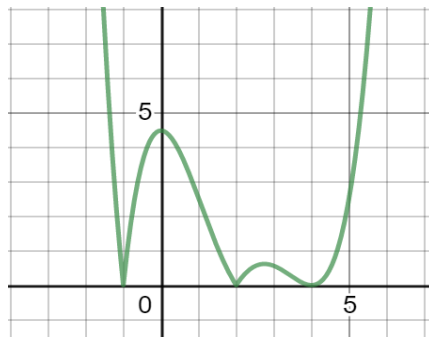
D.  $\sin 2x$

5 The diagram shows the graph of the function  $y = f(x)$ .

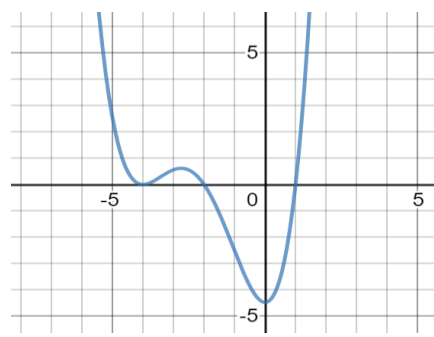


Which graph represents  $y = f(|x|)$ ?

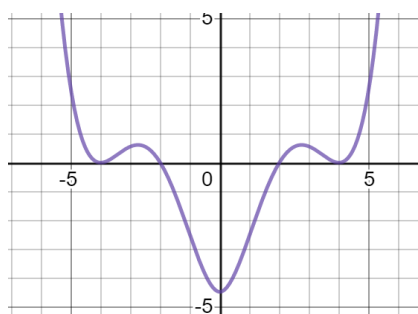
A.



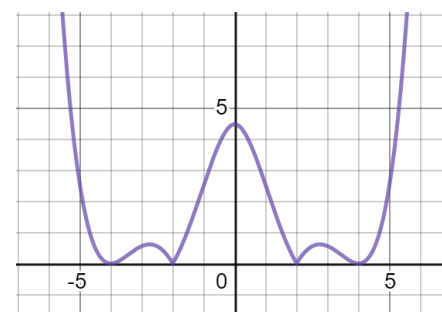
B.



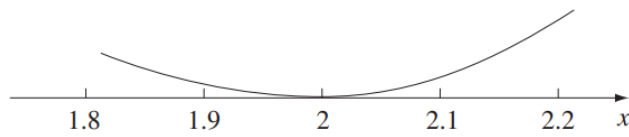
C.



D.



- 6 Part of the graph of  $y = P(x)$ , where  $P(x)$  is a polynomial of degree four, is shown below.



- Which of the following could be the polynomial of  $P(x)$ ?
- A.  $P(x) = x^2(x+2)^2$
- B.  $P(x) = (x+2)^4$
- C.  $P(x) = x(x-2)^3$
- D.  $P(x) = (x-1)^2(x-2)^2$
- 7 How many arrangements of the letters of the word OLYMPIC are possible if the M and C are to be together in any order?
- A.  $5!$
- B.  $6!$
- C.  $2 \times 5!$
- D.  $2 \times 6!$
- 8 When the polynomial  $P(x)$  is divided by  $(x+1)(x-3)$ , the remainder is  $2x + 7$ .

What is the remainder when  $P(x)$  is divided by  $(x-3)$ ?

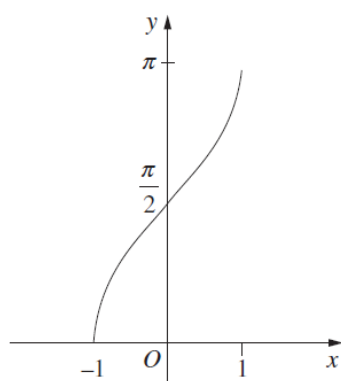
- A. 13
- B. 9
- C. 7
- D. 1

- 9 A circle's circumference is given by  $C = \pi D$ . Its area is given by  $A = \pi r^2$ , where  $D$  is the diameter and  $r$  is the radius.

The expression for  $\frac{dA}{dC}$  is given by:

- A.  $\frac{1}{r}$
- B.  $r$
- C.  $\frac{1}{2\pi}$
- D.  $2\pi$

- 10 The diagram shows the graph of a function.



Which function does the graph represent?

- A.  $y = \cos^{-1} x$
- B.  $y = -\cos^{-1} x$
- C.  $y = \frac{\pi}{2} + \sin^{-1} x$
- D.  $y = -\frac{\pi}{2} - \sin^{-1} x$

**End of Section I**

## Section II

**60 marks**

**Attempt Questions 11 – 14**

**Allow about 1 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use a SEPARATE writing booklet.

- (a) Solve  $|2x - 6| \geq 8$ . **2**
- (b) Write  $\frac{1}{n!} - \frac{1}{(n+1)!}$  as a single fraction. **1**
- (c) Forty-five balls, numbered 1 to 45, are placed in a barrel, and one ball is drawn at random. What is the probability that the number on the ball drawn is odd? **1**
- (d) In how many ways can a committee of 4 men and 5 women be selected from a group of 8 men and 9 women? **1**
- (e) The polynomial  $P(x) = x^2 + ax + b$  has a zero at  $x = 2$ . When  $P(x)$  is divided by  $x + 1$ , the remainder is 18. **3**
- Find the value of  $a$  and  $b$ .
- (f) A function  $f(x)$  is given by  $x^2 - 4x + 3$ .
- (i) Explain why the domain of the function  $f(x)$  must be restricted if  $f(x)$  is to have an inverse function. **1**
- (ii) Give the equation for  $f^{-1}(x)$  if the domain of  $f(x)$  is restricted to  $x \geq 2$ . **2**
- (iii) State the domain and range of  $f^{-1}(x)$ , given the restriction in part (ii). **2**
- (iv) Sketch the curve  $y = f^{-1}(x)$ . **2**

**End of Question 11**

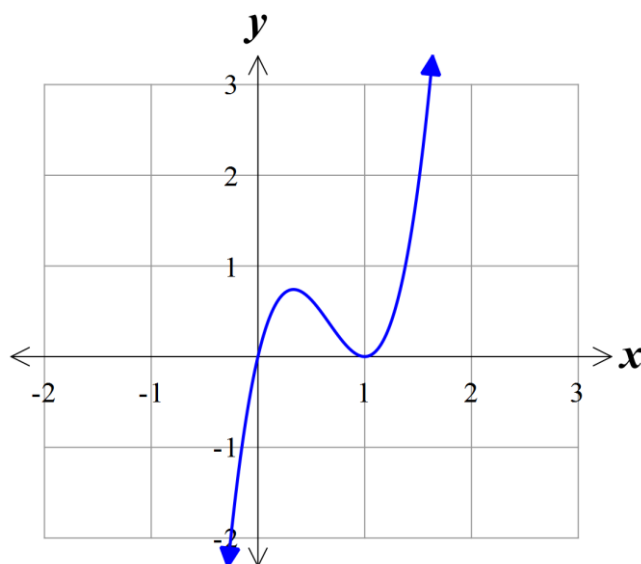
**Question 12** (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Find the domain and range of the function  $y = 2 \sin^{-1}(x-1)$ . 2
- (ii) Sketch the graph of the function. 1

- (b) Find the exact value of  $\cos^{-1}\left(\cos\left(-\frac{\pi}{4}\right)\right)$ . 1

- (c) The polynomial  $P(x) = x^3 - 12x + k$ , has a double zero. 3  
Find the two possible values of  $k$ .

- (d) The diagram shows the graph of  $y = f(x)$ .



By using the graph  $y = f(x)$  on the answer sheet provided, sketch the following functions. Your sketches should show all asymptotes and intercepts.

- (i)  $y = \frac{1}{f(x)}$  3

- (ii)  $y = \frac{1}{|f(x)|}$  1

**Question 12 continues on page 9**

Question 12 (continued)

- (e) A refrigerator has a constant temperature of  $3^{\circ}\text{C}$ . A can of drink with temperature  $30^{\circ}\text{C}$  is placed in the refrigerator.

After being in the refrigerator for 15 minutes, the temperature of the can of drink is  $28^{\circ}\text{C}$ .

The change in the temperature of the can of drink can be modelled by  $\frac{dT}{dt} = k(T - 3)$ , where  $T$  is the temperature of the can of drink,  $t$  is the time in minutes after the can is placed in the refrigerator and  $k$  is a constant.

- (i) Show that  $T = 3 + Ae^{kt}$ , where  $A$  is a constant, satisfies **1**  
 $\frac{dT}{dt} = k(T - 3)$ .
- (ii) After 60 minutes, at what rate is the temperature of the can of drink changing? **3**

**End of Question 12**

**Question 13** (15 marks) Use a SEPARATE writing booklet.**Marks**

- (a) A discrete random variable  $X$  has the probability distribution table shown. **3**

|         |     |     |     |     |
|---------|-----|-----|-----|-----|
| $X = x$ | 11  | 12  | 13  | 14  |
| $P(x)$  | 0.2 | 0.3 | $m$ | 0.4 |

By finding the value of  $m$ , calculate the expected value and the variance of  $X$ .

- (b) Show, without a calculator, that  $2 \cos 75^\circ \cos 15^\circ = \frac{1}{2}$ . **2**

- (c) Sixteen bananas are shared amongst five monkeys. Show that there is at least one monkey who receives at least four bananas. **1**

- (d) If one zero of the cubic  $f(x) = ax^3 + bx^2 + cx + d$  is the opposite of another, prove that  $ad = bc$ . **3**

- (e) A 10-metre ladder is leaning against a wall, and the base is sliding away from the wall at 7 cm/s. Find the rate at which the height is changing when the base of the ladder is 1.8 m from the wall, to 3 decimal places. **3**

- (f) By using  $t = \tan \frac{\theta}{2}$ , solve the equation  $\sqrt{3} \sin \theta + \cos \theta = 1$  for  $0 \leq \theta \leq 2\pi$ . **3**

**End of Question 13**

**Question 14** (15 marks) Use a SEPARATE writing booklet.

**Marks**

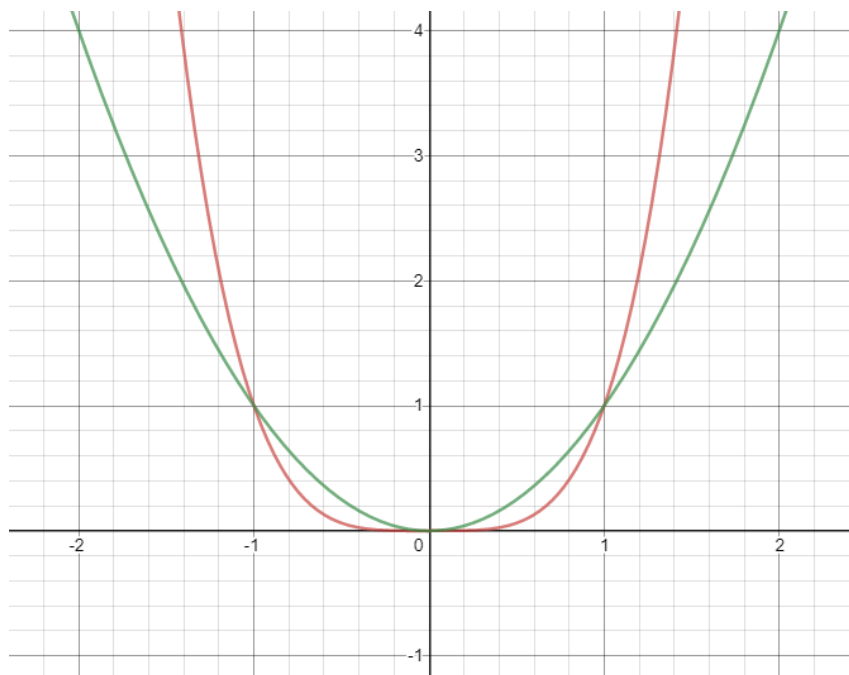
(a) Solve the inequation  $\frac{6-x^2}{x} \geq 1$ . 3

(b) (i) Prove the trigonometric identity  $\sin 3\theta = -4\sin^3 \theta + 3\sin \theta$ . 3

(ii) Hence find expressions for the exact values of the solutions to the equation  $8x^3 - 6x = -1$ . 4

(c) The diagram shows the graph of  $f(x) = x^4$  and  $g(x) = x^2$ . 2

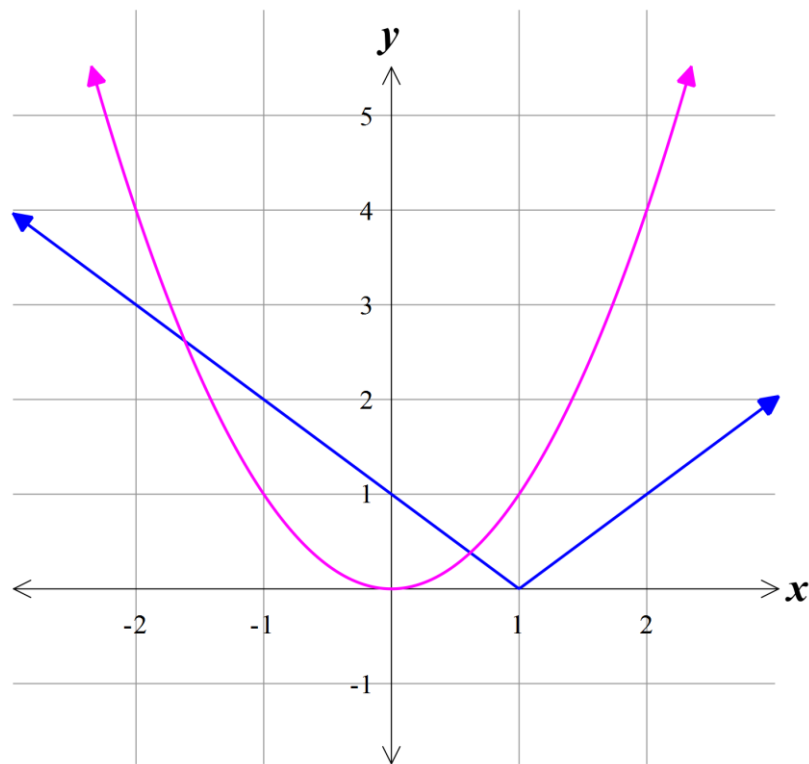
On the answer sheet provided, sketch  $y = f(x) - g(x)$ .



**Question 14 continues on page 12**

- (d) An absolute value function  $y = f(x)$ , and a parabolic function,  $y = g(x)$  are shown below.

3

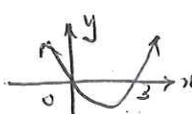


Use the answer sheet provided to sketch the graph of  $y = f(x) \times g(x)$ .

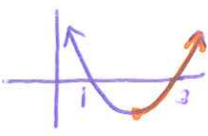
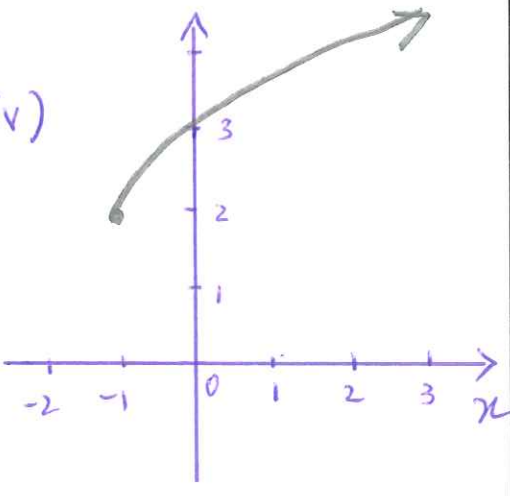
**End of Examination**



2020 Year 11 Mathematics Extension 1 Yearly SOLUTIONS

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Comments | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Comments |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><u>SECTION I</u></p> <p>1. <math>\log_a 2^3 = 1.893</math> (B)<br/> ie) <math>3 \log_a 2 = 1.893</math><br/> So, <math>\log_a 2 = 0.631</math><br/> <math>\therefore \log_a 4 = 1.262</math></p> <p>2. <math>x^2 - 3x \geq 0</math><br/> <math>x(x-3) \geq 0</math> <br/> <math>x \leq 0</math> or <math>x \geq 3</math> (C)</p> <p>3. For <math>ax^3 + bx^2 + cx + d</math><br/> <math>a = 1</math><br/> <math>\alpha + \beta + \gamma = -\frac{b}{a}</math><br/> ie) <math>-2 = -b</math><br/> <b><math>b = 2</math></b><br/> <math>\alpha\beta\gamma = -\frac{d}{a}</math> (B)<br/> ie) <math>1 = -d</math><br/> <b><math>d = -1</math></b></p> <p>4) <math>\sin 4x = 2 \sin 2x \cos 2x</math><br/> ie) <math>\sin 2x \cos 2x = \frac{1}{2} \sin 4x</math><br/> (A)</p> <p>5) (C)</p> |          | <p>6. (D) includes <math>(x-2)^2</math></p> <p>7. <br/> 2 ways (M &amp; C together)<br/> <math>6! \times 2</math> (D)</p> <p>8. <math>P(x) = (x+1)(x-3)Q(x) + 2x+7</math><br/> <math>P(3) = (3+1)(3-3)Q(3) + 2(3)+7</math><br/> <math>= 13</math> (A)</p> <p>9. <math>C = \pi D</math>      <math>A = \pi r^2</math><br/> <math>C = 2\pi r</math>      <math>\frac{dA}{dr} = 2\pi r</math><br/> <math>\frac{dC}{dr} = 2\pi</math><br/> <math>\therefore \frac{dA}{dC} = \frac{dA}{dr} \cdot \frac{dr}{dC}</math> (B)<br/> <math>= 2\pi r \times \frac{1}{2\pi}</math><br/> <math>= r</math></p> <p>10. <math>\sin^{-1} x</math> move up<br/> by <math>\frac{\pi}{2}</math>.<br/> (C)</p> |          |



| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Comments                                                                                               | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Comments                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <p><u>SECTION II</u></p> <p><u>Question 11</u></p> <p>(a) <math>2x - 6 \geq 8</math>      <math>-2x + 6 \geq 8</math><br/> <math>2x \geq 14</math>      <math>-2x \geq 2</math><br/> <math>x \geq 7</math>      or      <math>x \leq -1</math></p> <p>(b) <math>\frac{(n+1) - 1}{n! (n+1)}</math><br/> <math>= \frac{n}{(n+1)!}</math></p> <p>(c) <math>\frac{23}{45}</math></p> <p>(d) <math>{}^8C_4 {}^9C_5 = 8820</math></p> <p>(e) <math>P(x) = x^2 + ax + b</math><br/> <math>P(2) = 0</math></p> <p>(ie) <math>2^2 + a(2) + b = 0</math><br/> <math>2a + b = -4</math> — ①</p> <p><math>P(-1) = 18</math></p> <p>(ie) <math>(-1)^2 + a(-1) + b = 18</math><br/> <math>-a + b = 17</math> — ②</p> <p>Solve ① &amp; ② simultaneously</p> <p><math>3a = -21</math><br/> <math>\therefore a = -7</math>      <math>b = 10</math></p> | <p>Also accept<br/> <math>\frac{n}{(n+1)n!}</math><br/> or<br/> <math>\frac{1}{(n+1)(n-1)!}</math></p> | <p>(f) <math>f(x) = x^2 - 4x + 3</math><br/> <math>= (x-3)(x-1)</math></p> <p> • It fails horizontal line test.<br/> • Not one-to-one function</p> <p>(ii) <math>f^{-1}(x): x = y^2 - 4y + 3</math><br/> <math>x = y^2 - 4y + 4 - 1</math><br/> <math>x + 1 = (y - 2)^2</math> } ①<br/> <math>y - 2 = \pm \sqrt{x + 1}</math><br/> <math>y = 2 \pm \sqrt{x + 1}</math> } <math>y \geq 2</math><br/> So <math>f^{-1}(x): y = 2 + \sqrt{x + 1}</math> ①</p> <p>iii) Domain <math>x \geq -1</math><br/> Range <math>y \geq 2</math></p> <p>iv) </p> | <p>Domain of <math>f(x)</math><br/> <math>x \geq 2</math><br/> Range of <math>f(x)</math><br/> <math>y \geq -1</math></p> |

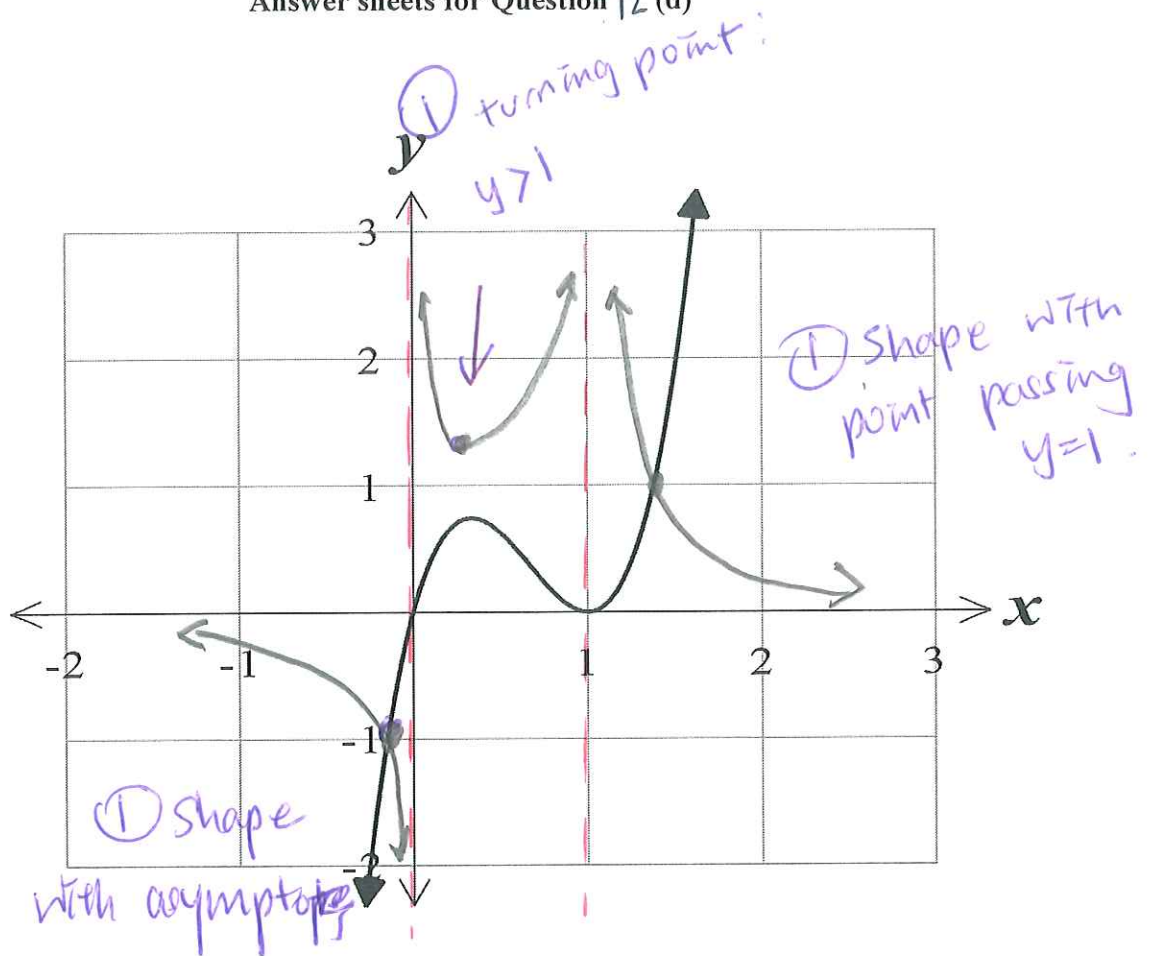


# 2020 Year 11 Mathematics Extension 1 Yearly SOLUTIONS

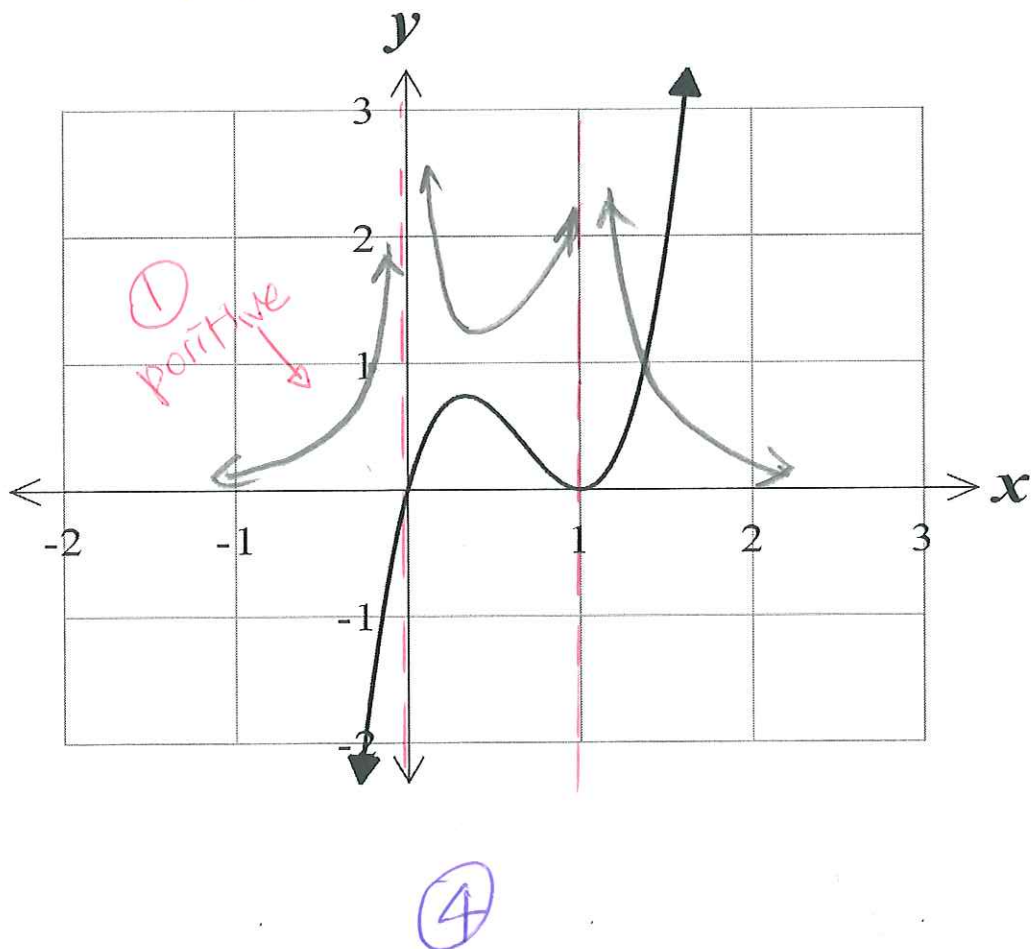
| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Comments                                          | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Comments                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <p><u>Question 12</u></p> <p>a) (i) <math>\frac{y}{2} = \sin^{-1}(x-1)</math><br/> Domain <math>-1 \leq x-1 \leq 1</math><br/> ie) <math>0 \leq x \leq 2</math><br/> Range <math>-\frac{\pi}{2} \leq \frac{y}{2} \leq \frac{\pi}{2}</math><br/> ie) <math>-\pi \leq y \leq \pi</math></p> <p>(ii) </p> <p>(b) <math>\cos(-\frac{\pi}{4}) = \frac{1}{\sqrt{2}}</math><br/> Let <math>\alpha = \cos^{-1}(\frac{1}{\sqrt{2}})</math><br/> <math>\cos \alpha = \frac{1}{\sqrt{2}}</math><br/> <math>\alpha = \frac{\pi}{4}</math><br/> So <math>\cos^{-1}(\cos(-\frac{\pi}{4})) = \frac{\pi}{4}</math></p> <p>(c) <math>P(x) = x^3 - 12x + k</math><br/> <math>P'(x) = 3x^2 - 12</math><br/> Let the root be <math>\alpha</math><br/> <math>P'(\alpha) = 0</math><br/> ie) <math>3\alpha^2 - 12 = 0</math><br/> <math>\alpha^2 = 4</math><br/> <math>\therefore \alpha = \pm 2</math> ①</p> | <p>✓ shape</p> <p>(e.c) from part(i)</p> <p>③</p> | <p>(c) wht. <math>P(2) = 0</math><br/> <math>P(2) \Rightarrow 2^3 - 12(2) + k = 0</math><br/> ie) <math>k = 16</math> ①</p> <p><math>P(-2) \Rightarrow (-2)^3 - 12(-2) + k = 0</math><br/> <math>k = -16</math> ①</p> <p>(d) see separate diagram</p> <p>(e)(i) <math>T = 3 + Ae^{kt}</math><br/> <math>\frac{dT}{dt} = k \cdot Ae^{kt}</math><br/> wht <math>Ae^{kt} = T - 3</math><br/> <math>\therefore \frac{dT}{dt} = k(T - 3)</math></p> <p>(ii) <math>t = 60</math> <math>\frac{dT}{dt} = ?</math><br/> <math>t = 0</math> <math>T = 30</math><br/> ie) <math>T = 3 + Ae^{kt}</math><br/> <math>30 = 3 + Ae^0</math><br/> <math>\therefore A = 27</math> ① for A</p> <p><math>t = 15</math> <math>T = 28</math><br/> ie) <math>28 = 3 + 27e^{15k}</math><br/> <math>\frac{25}{27} = e^{15k}</math><br/> <math>k = \frac{1}{15} \ln(\frac{25}{27})</math> ① for k</p> <p><math>t = 60</math>, <math>T = 3 + 27e^{k \cdot 60}</math><br/> <math>= 22.8458...</math></p> <p>So <math>\frac{dT}{dt} = \frac{1}{15} \ln(\frac{25}{27}) (22.8458 - 3)</math><br/> <math>= -0.1018</math> ①</p> <p>Temp. is decreasing at <math>0.102^\circ/\text{min}</math></p> | <p><math>k = -0.00513</math></p> |

Answer sheets for Question 12 (d)

(i)



(ii)





## Suggested Solution (s)

## Comments

Question 13

(a)

| $x$      | 11   | 12   | 13   | 14   | Sum   |
|----------|------|------|------|------|-------|
| $P(x)$   | 0.2  | 0.3  | 0.1  | 0.4  | 1     |
| $E(x)$   | 2.2  | 3.6  | 1.3  | 5.6  | 12.7  |
| $E(x^2)$ | 24.2 | 43.2 | 16.9 | 78.4 | 162.7 |

$$m = 0.1 \quad \checkmark$$

$$E(x) = 12.7 \quad \checkmark$$

$$\begin{aligned} \text{Var}(x) &= E(x^2) - m^2 \\ &= 162.7 - (12.7)^2 \\ &= 1.41 \quad \checkmark \end{aligned}$$

(b) R.T.P  $2 \cos 75^\circ \cos 15^\circ = \frac{1}{2}$ 

$$\text{LHS} = 2 \cos 75^\circ \cos 15^\circ$$

$$= \cos(75^\circ + 15^\circ) + \cos(75^\circ - 15^\circ) \quad \checkmark$$

$$= \cos 90^\circ + \cos 60^\circ$$

$$= 0 + \frac{1}{2} \quad \checkmark$$

$$= \frac{1}{2}$$

$$= \text{RHS}$$

(c) There are 16 bananas &amp; 5 monkeys.

$$\text{ie) } \frac{16}{5} = 3\frac{1}{5} \quad \checkmark \quad (x \geq 3\frac{1}{5} = 4)$$

By the pigeonhole principle, there are at least one

Monkey receives at least 4 bananas.

(d) Let the roots be  $\alpha, -\alpha$  &  $\beta$ .

$$\alpha + (-\alpha) + \beta = -\frac{b}{a}$$

$$\text{ie) } \boxed{\beta = -\frac{b}{a}} \quad \text{--- (1)} \quad \checkmark$$

$$-\alpha^2 + \alpha\beta - \alpha\beta = \frac{c}{a}$$

$$-\alpha^2 = \frac{c}{a}$$

$$\text{ie) } \boxed{\alpha^2 = -\frac{c}{a}} \quad \text{--- (2)} \quad \checkmark$$

$$\alpha(-\alpha)\beta = -\frac{d}{a}$$

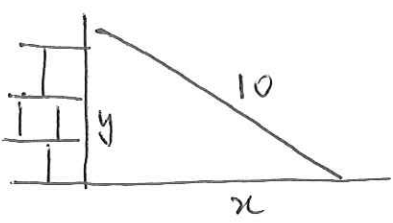
$$\text{ie) } \boxed{-\alpha^2\beta = -\frac{d}{a}} \quad \text{--- (3)} \quad \checkmark$$

$$\therefore -\left(-\frac{c}{a}\right)\left(-\frac{b}{a}\right) = -\frac{d}{a}$$

$$\therefore bc = da \quad \checkmark$$



## 2020 Year 11 Mathematics Extension 1 Yearly SOLUTIONS

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Comments                                                   | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Comments                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <p>Q13 cont.</p> <p>(e) </p> $\frac{dx}{dt} = 7 \text{ cm/s}$ $= 0.07 \text{ m/s}$ <p><math>\frac{dy}{dt} = ?</math> when <math>x = 1.8 \text{ m}</math></p> <p>Now <math>x^2 + y^2 = 10^2</math></p> <p>(e) <math>y^2 = 100 - x^2</math></p> $y = (100 - x^2)^{\frac{1}{2}}$ $\frac{dy}{dx} = \frac{1}{2}(100 - x^2)^{-\frac{1}{2}} \times -2x$ $= \frac{-x}{\sqrt{100 - x^2}}$ <p><math>\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}</math> @ <math>x = 1.8 \text{ m}</math></p> $= \frac{-1.8}{\sqrt{100 - 1.8^2}} \times 0.07$ $= -0.0128092 \dots$ $\approx -0.013 \text{ m/s}$ <p>OR decreasing at <math>1.28 \text{ cm/s}</math></p> | <p>(1)</p> <p>(1)</p> <p>(1) sign &amp; unit included.</p> | <p>(f) Let <math>t = \tan \frac{\theta}{2}</math></p> $\sqrt{3} \sin \theta + \cos \theta = 1$ <p>(e) <math>\frac{2\sqrt{3}t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1</math></p> $2\sqrt{3}t - t^2 = 1 + t^2$ $2t^2 - 2\sqrt{3}t = 0$ $2t(t - \sqrt{3}) = 0$ <p><math>t = 0</math> or <math>t = \sqrt{3}</math></p> $\tan \frac{x}{2} = 0$ $\tan \frac{x}{2} = \sqrt{3}$ <p><math>\frac{x}{2} = 0, \pi</math> or <math>\frac{x}{2} = \frac{\pi}{3}</math></p> <p><math>\therefore x = 0, 2\pi</math> or <math>\frac{2\pi}{3}</math></p> | <p>(1)</p> <p>(1)</p> <p>(1)</p> |

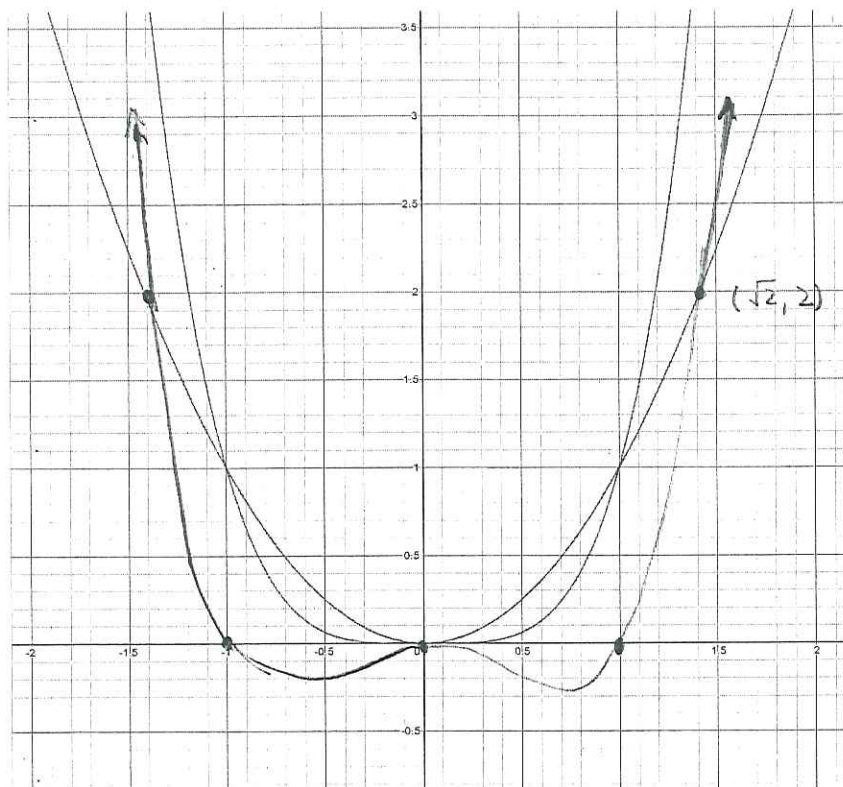
⑥



# 2020 Year 11 Mathematics Extension 1 Yearly SOLUTIONS

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Comments                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <p><u>Question 14</u></p> <p>a) <math>x^2 \times \frac{6-x^2}{x} \geq 1 \times x^2</math> note <math>x \neq 0</math></p> $x(6-x^2) \geq x^2 \quad (1)$ $x(6-x^2) - x^2 \geq 0$ $-x[6+x^2+x] \geq 0$ $-x(x+3)(x-2) \geq 0 \quad (1)$ <p>ie <math>x(x+3)(x-2) \leq 0</math></p> <p><math>\therefore x \leq -3</math> or <math>0 &lt; x \leq 2 \quad (1)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                               |
| <p>(b) <math>\sin(2\theta + \theta) = \sin 2\theta \cos \theta + \cos 2\theta \sin \theta \quad (1)</math></p> <p>(i)</p> $= (2\sin \theta \cos \theta) \cos \theta + (2\cos^2 \theta - 1) \sin \theta \quad (1)$ $= 2\sin \theta \cos^2 \theta + 2\sin \theta \cos^2 \theta - \sin \theta$ $= 4\sin \theta (1 - \sin^2 \theta) - \sin \theta \quad (1)$ $= 4\sin \theta - 4\sin^3 \theta - \sin \theta$ $\therefore \sin 3\theta = 3\sin \theta - 4\sin^3 \theta$                                                                                                                                                                                                                                                                                                                        |                                                               |
| <p>(b)(ii) <math>8x^3 - 6x = -1</math></p> $-2(3x - 4x^3) = -1$ $3x - 4x^3 = \frac{1}{2}$ <p>Let <math>x = \sin \theta</math></p> $4\sin^3 \theta - 3\sin \theta = -\sin 3\theta$ $8\sin^3 \theta - 6\sin \theta = -2\sin 3\theta$ <p><math>\therefore \sin 3\theta = \frac{1}{2} \quad (1)</math></p> <p><math>3\theta = \frac{\pi}{6}, \pi - \frac{\pi}{6}, 2\pi + \frac{\pi}{6}, 3\pi - \frac{\pi}{6}, 4\pi + \frac{\pi}{6}, \dots</math></p> <p><math>\theta = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}, \dots</math></p> <p><math>\therefore x = \sin\left(\frac{\pi}{18}\right), \sin\left(\frac{5\pi}{18}\right), \sin\left(\frac{25\pi}{18}\right) \quad (1)</math></p> <p>or <math>-\sin\frac{\pi}{18}</math></p> | <p><math>x \neq 0</math></p> <p>for 3 distinct solutions.</p> |

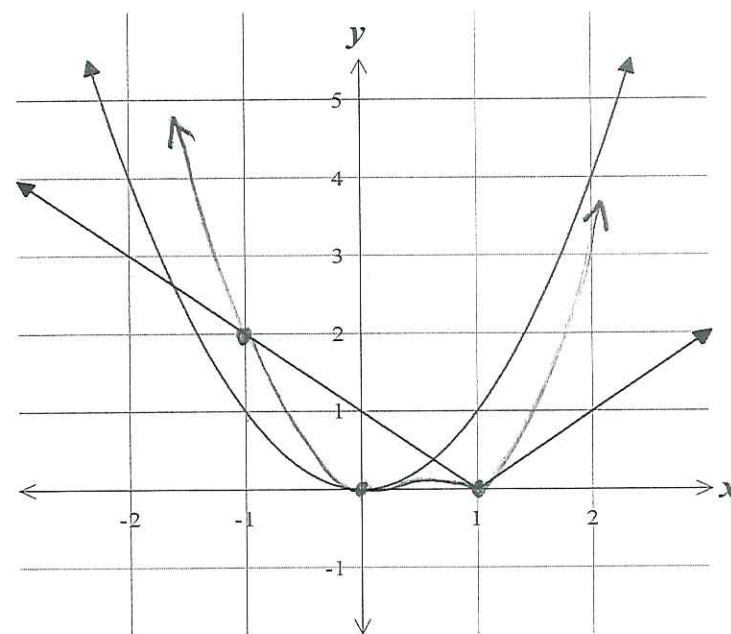
Answer sheet Q 14 (c)



- ① 3  $x$  intercepts
- ① shape including behaviour as  $x \rightarrow \infty$

⑧

Answer sheet Q 14 (d)



- ✓ 2  $x$ -Intercepts
- ✓ passing  $(-1, 2)$  (or a point)
- ✓ shape

⑨